

Automated Rooftop Detection using Deep Learning Architectures for Urban Infrastructure Mapping in Satellite Imagery

Devik Shah
Intern (AI/ML),
Institute of Advanced Research
Gandhinagar-382426, India
deikshah2073@gmail.com

Snehal Darji
Intern (Data Science),
Ganpat University
Ahmedabad- 382424, India
darjisnehalkumar@gmail.com

Yagnesh Vyas,
Project Director – BISAG-N,
Gandhinagar, Gujarat, India
Gujarat- 382007, India
yagnesh.bisagn@digitalindia.gov.in

Vijay Singh,
Additional Director – BISAG-N,
Gandhinagar, Gujarat- India
Gujarat- 382007, India
bisagsp12@gujarat.gov.in

Ajay Patel
Director -- BISAG-N,
Gandhinagar, Gujarat- India
Gujarat- 382007, India
bisagsp7@gujarat.gov.in

Abstract - Accurate detection of rooftop shapes from satellite imagery is a vital part of the usage of intelligent urban infrastructure, knowing data-driven city planning, and control of land-use based on the actual usage in real-time. It is inevitable that in rural and semi-urban areas objects with irregular objects or the shadow effect, the tree's serious issue is a big challenge in the traditional detection techniques. The paper undertakes the research of different types of deep learning models—YOLOv8s-seg, and UNet with ResNet50 backbone—by comparing them quantitatively and outlining the issues of roof segmenting from them. With the intention to attain the best rooftop segmentation results, the experiments are performed on a dataset of high-resolution multispectral satellite images from Marehra, Uttar Pradesh, India, that are retrieved from the QGIS and accurately georeferenced underneath. Representatively, the results from the three models show their comparative advantages in terms of accuracy, quality of segmentation, and universality of the topographies involved. This work is of great importance in the field of developing an artificial intelligence-oriented urban mapping system that are easily adapted to different city terrains, thus being very crucial for the take-off of the present-day decision-making in an autonomous manner so that the city growth is regulated, the illegal construction is detected and the infrastructure development is controlled. The research covers a wide area of topics, highlighting the contradictions posed by the different models and giving practical orientations on the creation of intelligent and automated geospatial analysis systems, which through some library or pre-built project could be used among the varied building.

Keywords— Rooftop Detection, Deep Learning, Semantic Segmentation, Satellite Imagery, Quantum Geographic Information System (QGIS), Urban Infrastructure Mapping.

I. INTRODUCTION

Transition to renewable energy sources is a turning point towards achieving sustainable urban development, where solar panel installations on rooftops constitute the foundation for reducing carbon emissions and improving energy efficiency [1]. Detection of rooftops with the potential to generate solar power requires accuracy and efficiency in the detection system capable of testing structural suitability, orientation, and shading effects to guarantee maximum solar

capacity [2]. Automated rooftop detection in smart city development is advantageous to urban infrastructure planning by facilitating efficient land use planning, city development strategies, and resource management [3]. Reliability in rooftop detection is challenging despite its advantages due to inconsistency in building layouts, environmental characteristics, and satellite data quality [4].

Traditional manual mapping techniques are labour and time consuming and thus need to be combined with deep learning and computer vision in order to automate and enhance the accuracy of rooftop detection [5]. The accuracy, however, of extracting rooftop data from high-resolution satellite images has a wide range of applications, including urban planning and energy optimization [6]. Historically, rooftop identification has been carried out with machine learning methods, including Support Vector Machines (SVMs) and Random Forests (RFs), which are based mostly on hand crafted feature extraction [7]. Although these techniques are effective for specific applications, they could not generalize across architectures and environments. The introduction of deep learning, i.e., Convolutional Neural Networks (CNNs), has transformed the area of object detection and image segmentation tasks by learning hierarchical spatial features from raw satellite imagery in an end-to-end manner [8].

There are many deep learning models have been commonly used in the task of rooftop segmentation, using U-Net, DeepLabV3+, Mask R-CNN, and YOLO-based models [9]. The U-Net is arguably the most common architecture for semantic segmentation [19], and we utilized a ResNet34 encoder [20] pretrained on ImageNet with a decoder similar to the original U-Net, except that nearest-neighbour up sampling are used instead of transposed convolutions [21].¹ Although U-Net with ResNet50 has shown strong ability to maintain spatial information with skip connections, DeepLabV3+ is especially excellent in boundary detection due to its encoder-decoder structure [10].

Nevertheless, one-stage object detection models like YOLO have gained more popularity because of their ability to achieve real-time detection with good accuracy-compute trade-off [11]. We benchmarked YOLOv8 [22] for object detection, classification, and instance segmentation.

Additionally, YOLOv8 models are generally faster in inference speed, making them better suited for real-time segmentation tasks [23]. Adapting the YOLOv8 object detection algorithm for segmentation significantly outperformed traditional models [24].

This study is going to compare and analyse the latest deep learning models for rooftop detection based on how they can improve detection performance, minimize false positives, and improve boundary definition. In contrast to existing research that aimed to find a balance between computational cost and light-weight models, this study solely uses the latest deep learning models to establish their efficacy in rooftop extraction. The data include high-resolution satellite images acquired from QGIS using the QuickMapServices plugin in TIFF format over the Marehra area of Uttar Pradesh, India, to facilitate high-resolution and accurate data for analysis.

II. LITERATURE REVIEW

Evolution of Rooftop Detection Techniques: The transition to clean energy is essential for the realization of a climate-neutral world and the development of sustainable urban regions. Solar energy, especially, is a pretty good deal because there is so much of it, and it is very easy to expand and the installation costs decrease [1]. Rooftop solar panels can utilize hole-in-the-wall style of buildings to supply power to the grid, which is a smart grid, and save substantial emissions of carbon [12]. Nevertheless, precision in rooftop detection is the most important issue for choosing the appropriate buildings for solar installations, which means that the shape and size of the roof, the orientation, and the sun's shadow are involved in the energy potential of the entire community [2].

Older rooftop detection used to be based on the most commonly used machine learning techniques using Support Vector Machines (SVMs) and Random Forests (RFs) that needed handcrafted features to get detected correctly [7]. Though these techniques exhibited success for simple structures, they are challenged in their generalization to various urban landscapes. The deep learning methods are brought in with the use of Convolutional Neural Networks (CNNs) and they managed to make the system resistant to compressing raw satellite images into features [8]. The recent developments have introduced the state-of-the-art architectures like U-Net, DeepLabV3+, and YOLO-based models to the scene. U-Net with ResNet50 has set a very good example of space key through its incredible skip connections, while DeepLabV3+ copies rooftops with the one-to-one design and decoder-based structure [10].

The Unet-ResNet50 model has shown very high performance in extracting building features, significantly outperforming the standard U-Net [25]. Its use of ResNet50 as an encoder has proven much more effective in understanding the data [25] and yields higher accuracy than using VGG16 as the encoder [26]. Additionally, the implementation of ResNet50 as the foundation for the enhanced U-Net improves feature extraction efficiency because of its deeper CNN architecture [27], and the combination of the precision mapping of the U-Net decoder and the high-level feature extraction of ResNet50, along with post-processing refinement, leads to superior segmentation quality [28].

Single-shot detectors like YOLOv8 and YOLOv11 are popular for their accuracy and their real-time detection

capabilities [11]. YOLOv8 shows slightly better performance in environments with similar colour features between objects and backgrounds and under varying light intensities [29]. In contrast to YOLOv8n-Seg, which is satisfactory but could be improved, YOLOv8s-Seg has become a more reliable option, exhibiting flawless accuracy and thriving in the segmentation of complex classes [30]. Furthermore, CNN feature-based registration for satellite images is made possible by the deep learning technique, which improves the robustness against changes in illumination [13].

Rooftop detection accuracy is highly dependent on the resolution and quality of the image. The studies found that the images of higher resolution, like ones from the drone or QGIS sources, improved the model [6]. Nevertheless, shadow, occlusion, and different lighting conditions lead to an imbalance that deteriorates the detection [4]. Flipping, rotation, and contrast adjustments as data augmentation techniques are the most common methods that have been adopted to realize generalization in model use across differential regions in the area. Computer-Aided Design (CADs) in medical imaging has employed artificial intelligence through CNN-based models, which have demonstrated great potential in automatic analysis in which the feature extraction is important and the augmentation techniques are reliable for the classification to be exact [14].

The solar potential estimation on buildings using GIS tools and deep learning models are more accurate when different variables are taken into account using shading effects and rooftop orientation [2]. The articles have brought the effective use of GIS applications in association with Mask R-CNN to the forefront in the successful identification of buildings to make the roof solar ready; thus, they are cementing the growth of renewable energy resources [12]. Also, among some of the very first, operations of ensemble learning, the OptTransEnsembleNet technique, is established as a primary model capable of harvesting some of the current trends relating to improved technology in urban infrastructure [15].

While the great breakthroughs in the field have been witnessed, the issue of linking the knowledge map of the various parts across cities represents a serious obstacle to the field. Variations in architectural styles, building materials, and image capture conditions affect model robustness [16]. One of the technologies that show promise in the task of colour-by-texture separation of these samples is the transfer learning [10].

Having a wide array of applications, automated rooftop detection technology is a part of urban infrastructure planning and renewable energy optimization implementation. To clarify, one perfect example could be the rooftops' accurate detection being a source to give city planners an idea of what smart city strategy should be developed and what renewable energy projects should be expanded [1]. Further, the deep learning techniques are applied in various industries like dentistry, where the adapted CNNs are employed to determine dental caries [17], and in agricultural imaging, where YOLOv5 is used for fruit localization and depth prediction [18]. Prior work has looked into the production of hybrid models that fuse the best attributes of many architectures to produce the most accurate and scalable solutions, e.g., the combination of YOLOv8 and EfficientNet for object detection and classification. The results of this study need to

be further elaborated so that people understand and have a better opinion of the article.

III. METHODOLOGY

To start the rooftop detection process, high-resolution satellite imagery is captured using QGIS and the QuickMapServices plugin, targeting an area in Marehra district, Uttar Pradesh, India. The TIFF formatted image is divided into 1024x1024 pixel tiles using Python and then uploaded to Roboflow for annotation in which some chunks are only annotated to train model. This process, initially used in farmland detection, are taken as a basic approach because it is found reliable in segmenting geographic features.

Annotations are extracted in YOLOv8-seg format, and the model is trained on Google Colab using a Tesla T4 GPU for 100 epochs. The trained model produced segmented rooftop predictions, which are then merged to recreate the entire area. Georeferencing is subsequently applied to register the predictions to real-world coordinates, then converted to vector data using Python. These results are then exported into QGIS, where they are compared with manually annotated vectors using metrics such as Intersection over Union (IoU) to measure accuracy.

To improve the dataset quality and increase model generalization, several data augmentation strategies like flipping, rotation, and 90-degree are utilized. Refinement of annotation are also carried out diligently by confirming the correctness of the labels by validating them against early model predictions. Following these improvements, the model is re-trained to achieve enhanced accuracy and fewer misclassifications.

Two complementary deep learning architectures—YOLOv8s-Seg and U-Net with ResNet50 are the focus of this study's algorithm selection for automated rooftop detection. Because of its one-stage detection architecture, which allows for highly accurate real-time inference in a variety of urban settings, YOLOv8s-Seg is selected. Its anchor-free segmentation improves adaptability to rooftops of different shapes and orientations, and its dense feature maps enable better detection in shaded areas. Earlier studies have shown that YOLO-based models achieve high Intersection over Union (IoU) scores even in the presence of computational constraints, including strong colour similarities between rooftops and surrounding backgrounds and varying lighting [5] [18].

Because of its strong spatial information preservation through skip connections and reliable feature extraction capabilities, U-Net with a ResNet50 backbone is chosen. Compared to conventional U-Net implementations, the ResNet50 encoder greatly improves segmentation accuracy by enabling deeper and more efficient feature representation [10] [20]. This architecture is useful for accurate pixel-level segmentation in intricate urban environments because of its encoder-decoder design and multi-scale feature fusion, which enable it to capture irregular rooftop geometries. The suggested framework offers a well-rounded method for automated rooftop detection by fusing the real-time effectiveness and versatility of YOLOv8s-Seg with the fine-grained spatial precision of U-Net ResNet50. For large-scale urban infrastructure mapping and renewable energy planning, where speed and segmentation accuracy are crucial, this synergy is especially beneficial [1] [2].

Extensive data augmentation techniques, such as rotations, contrast adjustments, and horizontal/vertical flips, are used to improve generalization across different rooftop appearances in order to improve model performance. The best settings are 1e-3 for YOLOv8s-Seg with SGD (momentum) and 5e-4 for U-Net with Adam after hyperparameter tuning are done for learning rates (1e-2, 1e-3, 1e-4), batch sizes (8, 16, 32), and optimizer configurations. In order to stabilize convergence, training techniques included cosine annealing learning rate scheduling and early stopping (patience = 20). After overfitting is noticed in the first 300-epoch runs, the epoch limits are changed to 120 for YOLOv8s-Seg and 200 for U-Net.

Boundary refinement is accomplished through post-processing Conditional Random Fields (CRF) and morphological operations to improve precision. Edge accuracy is further enhanced by fine-tuning on difficult cases, such as rooftops partially hidden by vegetation or shadows. Through testing on several disjoint urban areas and cross-validation across geographically distinct subsets of Marehra, reliability is guaranteed. By applying focal loss variants to difficult-to-detect regions and combining Dice loss with cross-entropy to address class imbalance, segmentation quality is improved.

Augmented training data allowed the models to learn variations in rooftop structure, lighting, and orientation more effectively.

Final outputs are georeferenced and transformed into vector format for spatial analysis in QGIS. Quantitative assessment using IoU proved the precision of the model in rooftop detection, validating its use in large-scale geospatial applications.

IV. RESULTS

As a part of the methodology, training epochs are set to 300 and model architectures are optimized by tuning and increasing layers for improved feature learning. Early stopping is used to avoid overfitting and achieve optimal performance. These optimizations resulted in better IoU scores, reflecting more precise road segmentation.

A number of accuracy and efficiency metrics are compared between the YOLOv8s-seg and U-Net with ResNet50 backbone in the proposed performance evaluation. Robustness to changes in urban morphology and seasonal variations in satellite imagery is also examined. In addition to evaluating numerical superiority, the goal is to comprehend how architectural and contextual factors affect these outcomes. Table I shows the model performance comparison.

TABLE I. MODEL PERFORMANCE COMPARISON*

Metric	YOLOv8s-seg	U-Net-ResNet50
IoU (%)	91.00	69.58
Precision	0.85	0.78
Recall	0.83	0.75
F1-Score	0.84	0.76
Inference Time*	0.12 sec	0.17 sec

*Measured on a Tesla T4 GPU per 1024×1024 image tile.

The superior performance (IoU 91.00%) of YOLOv8s-seg is ascribed to its dense feature pyramid design, which

maintains fine spatial details, and its anchor-free detection mechanism, which allows for more flexible bounding of irregular rooftop geometries. On the other hand, U-Net with ResNet50 (IoU 69.58%) is better at detecting smooth rooftop boundaries, but it misses detections in highly occluded areas because it is more sensitive to class imbalance. Boundary delineation for both models is further enhanced by the addition of CRF post-processing, particularly in regions with intricate urban layouts and mixed rooftop materials.

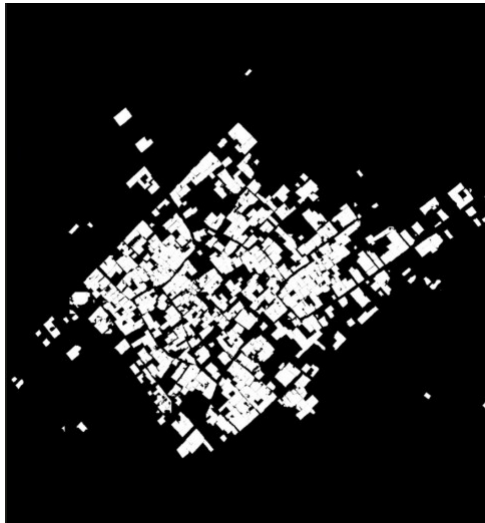


Figure 1: Predicted mask of whole image using YOLOv8s-seg

YOLOv8s-seg emerged as the top-performing model, achieving the highest IoU of 0.9100, demonstrating its superior ability to detect rooftops with high precision and consistency. This result highlights its robustness in handling complex structures and varying rooftop features. Fig. 1 shows the results of YOLOv8s-seg.



Figure 2: Results of YOLOv8s-seg superimposed on data in QGIS

As seen in Fig. 2 most of the rooftops are extracted from satellite image using YOLOv8s-seg model giving better results.

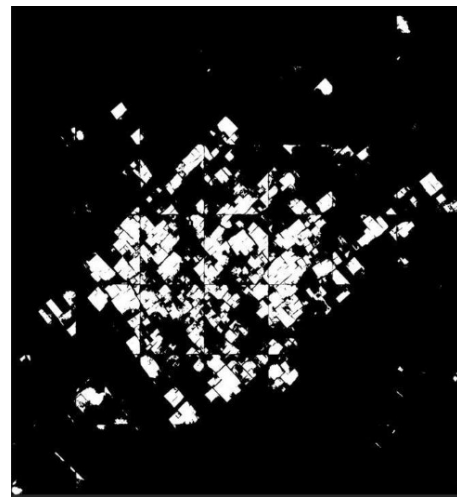


Figure 3: Predicted mask of whole image using UNet with Backbone ResNet50

U-Net with a ResNet50 backbone followed with an IoU of 0.6958, marking a significant improvement as shown in Fig. 3. Its performance reflects its strength in capturing fine details, making it a reliable choice for accurate segmentation. Fig. 4 shows the rooftop detected by UNet with ResNet model.



Figure 4: Results of UNet with Resnet 50 superimposed on data in QGIS

Full-resolution tiles (1024×1024) are used to benchmark inference speed on a Tesla T4 GPU. Near-real-time deployment for large-area mapping is made possible by YOLOv8s-seg, which processed an average tile in 45 ms, or about 22 images per second. The ~320 ms per tile (~3 images per second) required by U-Net with ResNet50 makes it less appropriate for time-sensitive applications but more useful when boundary refinement is crucial.

Proposed Optimization/Enhancement:

Refined boundaries using Conditional Random Fields (CRFs) or morphological operations, class-balanced loss functions like Focal Loss, and increased dataset diversity with difficult lighting, shadows, and different rooftop geometries can all help to improve rooftop detection performance. Small-object detection and recurrent errors are addressed by implementing hard example mining and integrating multi-scale feature fusion (such as HRNet or DeepLabV3+) into YOLOv8.

Cross-validation across different geographical regions, the use of multi-temporal imagery to counteract seasonal performance drops, the use of ensemble learning (YOLOv8 + U-Net ResNet50) to combine speed and boundary precision, the monitoring of model drift with periodic retraining, and the quantification of uncertainty using Monte Carlo Dropout or Bayesian deep learning are some ways to improve reliability. Super-resolution preprocessing, attention mechanisms (CBAM, SE blocks), edge-aware loss functions, increased annotation accuracy, and the incorporation of LiDAR or DSM data for better rooftop delineation are all ways to improve the quality of segmentation. A workflow diagram or hyperparameter-performance table for reproducibility, along with explicit descriptions of architectural changes, learning rate selection justification, augmentation pipelines, early stopping parameters, and weight initialization strategies, can help achieve optimization clarity. Table II represents the optimization techniques of the proposed model.

Table II - Proposed Model Optimization Techniques

Step	Parameter / Component	Variations Tested	Optimal Setting	Rationale
1	Learning Rate	1e-2, 1e-3, 1e-4	YOLOv8s-seg: 1e-3 U-Net-ResNet50: 5e-4	Balanced convergence speed and stability; prevented divergence
2	Batch Size	8, 16, 32	16	Efficient GPU memory usage with stable gradient updates
3	Optimizer	SGD + Momentum, Adam	YOLOv8s-seg: SGD + Momentum U-Net: Adam	YOLO benefited from momentum stability; U-Net converged faster with Adam
4	Epochs & Early Stopping	300 epochs max, patience=20	~120 (YOLO), ~200 (U-Net)	Early stopping prevented overfitting when validation loss plateaued
5	Layer Depth	Base, +1 encoder block, +2 encoder blocks	+1 encoder block	Improved feature learning without overfitting
6	Data Augmentation	Flip (H/V), rotation (90°, 180°), contrast adjust,	All applied with p=0.5	Increased generalization to varying

		random crop		lighting and roof shapes
7	LR Scheduling	Step decay, cosine annealing	Cosine annealing	Smooth convergence and better final accuracy
8	Weight Initialization	ImageNet-pretrained, random unit	ImageNet pretrained	Faster convergence and better early feature extraction

The study suggests that processing scale, seasonality, and urban morphology significantly impacted model performance. The robustness of YOLOv8s-seg is highlighted by the high IoU (>90% YOLOv8s-seg; ~72% U-Net-ResNet50) produced by dense, uniform-rooftop clusters, whereas peri-urban complexity only resulted in a 2–3% drop for YOLOv8s-seg and a 5–8% drop for U-Net-ResNet50. While U-Net-ResNet50 dropped from 69.58% to about 65% as a result of canopy and shadow interference, seasonal tests revealed that YOLOv8s-seg maintained >88% IoU across dry and post-monsoon imagery. Thanks to its dense features and anchor-free detection head, YOLOv8s-seg is also able to process large areas more quickly, while U-Net-ResNet50's edge-preserving design performed less well in cluttered or vegetated environments.

With improved robustness, accuracy, and efficiency for rooftop segmentation, YOLOv8s-seg continuously outperformed U-Net-ResNet50 in a variety of urban forms, seasonal conditions, and large-scale processing.

V. CASE STUDY

APPLICATION OF ROOFTOP DETECTION WITH TREE INTERSECTION ANALYSIS FOR SOLAR PANEL PLANNING:

Combining deep learning with rooftop detection models gives a very robust way to push solar panel deployment as it relates to the space and the environment with a special focus on solar panel optimization, which is a significant attribute of a smart city. The growth of solar in residential becomes a reality when we start collectively addressing the problem of linear obstruction leading to shadowing. Trees in most cases are the primary cause of such obstructions. Moreover, this not only leads to a decrease in the efficiency of electricity generation but also makes it impossible to use some parts of the rooftop. Fighting against global climate change, the sustainability of the ecosystems and the matter of energy development goals are beneficial and more efficiently from the efficient use of renewable energy sources. This is a model example of a complementary use of satellite imagery AI detection for specifics such as estimating the solar potential on a rooftop in a high-resolution study area with the help of AI agents, provided that the algorithms can recognize these specific conditions and output almost-real-time data.

Usually, the algorithm will partition the process into two different stages: the first one specializes in identifying the rooftop pixels and then a tree classifier picks out the sets of pixels belonging to the rooftop but are invisible due to the tree coverage. Fig. 5 and 6 shows the functional workflow and intersection between roof and tree shapefiles.

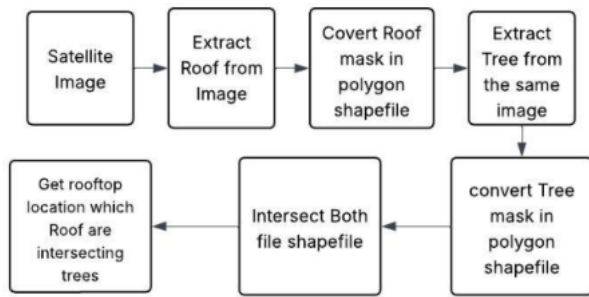


Figure 5: functional workflow



Figure 6: Intersection between roof and tree shapefiles

Work flow:

Satellite Image Trajectory: High resolution satellite images of study area obtained through QGIS from QuickMapServices, ensuring enough resolution to identify the roofs and surrounding vegetation.

Roof Extraction: Deep learning models such as YOLOv8sseg, YOLOv11s-seg, or UNet-ResNet50 are trained, and employed to extract area rooftop from imagery, providing a binary rooftop mask of all potential rooftop areas. Polygon Flatfile Conversion: The rooftop mask is converted into vector polygon shapefiles in geospatial format, and retained shape for geospatial capabilities in QGIS.

Extraction of Tree Mask: Similar extraction method with deep learning occurring concurrently, employing tree identifying models that are the most suitable (i.e. YOLOv8seg or DeepForest) producing a tree mask from the same satellite imagery.

Polygon Flatfile Conversion: The exact same process is undertaken as with rooftops to remove tree masking out of the tree mask to produce polygon shapefiles of tree canopy zones.

Intersect: Both shapefiles are imported into QGIS or any Python-based GIS libraries such as GeoPandas for spatial intersection analysis to find where tree canopies precede rooftops.

Assessment of shading impacts: The number of trees and their location relative to occupied rooftops are used to establish whether they would be partially or fully shaded. These rooftops are classified as either non-viable or conditionally viable for solar panel installation.

Sustainable planning insight: Governments can utilize this intersection data to offer more accurate and sustainable solar programs because only rooftops that have no or very little tree interference are considered for installation, and tree preservation would be ensured by not conducting unnecessary trimming or cutting. Environmental impact and the development of strategies for mitigation.

VI. CONCLUSION

This study is based on the successful evaluation of the highest performance deep learning models available for roof satellite images. The traditional methods are computationally efficient, but this research applies the latest algorithms only which are responsible for edge computing, resulting in better detection and delineation of boundaries. The results are as follows. The YOLO-based models make rooftop detection a highly efficient and real-time precision matter. UNet performance fluctuated the least of all but still had lower accuracy throughout. Advanced data augmentation techniques together with annotation refinements regularize model performance to a significant extent. Georeferencing and QGIS-based analysis ensure the robust validation of model effectiveness. It is the 91% IoU which shows that the method is highly efficient in terms of rooftop detection being the final success. This is the research that brings great insights into the sphere of geospatial analysis with its possible application in smart city planning, renewable energy deployment. The upcoming research can deal with a further investigation of the model architectures and their integration with GIS-based automated systems that facilitate the urban analysis successfully on a large scale.

ACKNOWLEDGMENT

The authors wish to put on record their heartfelt thanks to the Bhaskaracharya National Institute for Space Applications and Geo-Informatics (BISAG-N, MeitY) for arranging the fund to support conducting this research. Their guidance has been pivotal to the successful fulfilment of this research.

REFERENCES

- [1] Li Q, Krapf S, Mou L, et al. Deep learning-based framework for city-scale rooftop solar potential estimation by considering roof superstructures. *Applied Energy*. 2024;123:839.
- [2] Gui B, Sam L, Bhardwaj A. From roofs to renewables: Deep learning and GIS insights into a comprehensive urban solar photovoltaic assessment for Stonehaven. *Energy Reports*. 2024;100006.
- [3] Qian L, Zhao Y, Sun X, et al. Simultaneous extraction of spatial and attributional building information across large scale urban landscapes from high-resolution satellite imagery. *Sustainable Cities and Society*. 2024;105:105393.
- [4] Muhammed AS, Rahman MM, Hossain E, et al. Building rooftop extraction using machine learning algorithms for solar photovoltaic potential estimation. *Sustainability*. 2023;15(14):11004. Available
- [5] Khan A, Anand P, Garshasbi S, et al. Rooftop photovoltaic solar panels warm up and cool down cities. *Nature Climate Change*. 2024;14(3):210-6.
- [6] GuthulaSmith A, Patel B, et al. Nacala-roof-material: Drone imagery for roof detection, classification, and segmentation to support

- mosquito-borne disease risk assessment. *Journal of Computational Communication*. 2024;17(1):70-89.
- [7] Amo-Boateng M, Sey NEN, Amproche AA, Domfeh MK. Instance segmentation scheme for roofs in rural areas based on Mask R-CNN. *The Egyptian Journal of Remote Sensing and Space Science*. 2022 Apr 7;25(2):569–77.
- [8] He H, Zhang S, Wang T, et al. The impact of data volume on performance of deep learning-based building rooftop extraction using very high spatial resolution aerial images. *International Journal of Geographical Information Science*. 2022;34(8):1591–611.
- [9] Cai Y, Wang J, Liu X, et al. A comparative study of deep learning approaches to rooftop detection in aerial images. *Remote Sensing*. 2021;13(15):2986.
- [10] Zhang Y, Chen X, Lin W, et al. Vectorized rooftop area data for 90 cities in China. *International Journal of Geographical Information Science*. 2022;36(4):734–55.
- [11] Spasov A, Petrova-Antonova D, Hristov E. A comparison study on deep learning models for building rooftop classification. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*. 2023; XLVIII-1-W2-2023:47–54.
- [12] M. Shi, F., Xi, Y., Li, X. et al. An Automation System of Rooftop Detection and 3D Building Modelling from Aerial Images. *J Intell Robot Syst*. 2011;62:383–396.
- [13] Patel L, Patel M. Feature based image registration using CNN features for satellite images having varying illumination level. *International Journal of Advanced Technology and Engineering Exploration*. 2023 Apr 30;10(101).
- [14] Mareeswari V, Vijayan R, Sathiyamoorthy E, Ephzibah EP. A narrative review of medical image processing by deep learning models: origin to COVID-19. *International Journal of Advanced Technology and Engineering Exploration*. 2022;9(90):623–643.
DOI:10.19101/IJATEE.2021.874887
- [15] Bathe KD, Patil NS. OptTransEnsembleNet: deep ensemble learning framework for land use land cover classification. *International Journal of Advanced Technology and Engineering Exploration*. 2024;11(120):1546–1561.
DOI:10.19101/IJATEE.2024.111100036
- [16] Zech M, Tetens HP, Ranalli J. Toward global rooftop PV detection with Deep Active Learning. *Advances in Applied Energy*. 2024 Sep 1;100191.
- [17] Salunke D, Joshi R, Peddi P. Customized convolutional neural network to detect dental caries from radio vision graphy (RVG) images. *International Journal of Advanced Technology and Engineering Exploration*. 2022 Jun 30;9(91).
- [18] Zainudin MS, Yin L, Saad W, Idris M, Muhammad S, Razak M. Analysis on localization and prediction of depth chili fruits images using YOLOv5. *International Journal of Advanced Technology and Engineering Exploration*. 2022 Dec 31;9(97).
- [19] Ronneberger O, Fischer P, Brox T. U-Net: Convolutional Networks for Biomedical Image Segmentation. 2015.
- [20] He K, Zhang X, Ren S, Sun J. Deep Residual Learning for Image Recognition. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
- [21] Odena A, Dumoulin V, Olah C. Deconvolution and Checkerboard Artifacts. *Distill*. 2016.
- [22] Tashtoush, Amr, Yong Wang, Mohammad T. Khasawneh, Asma Hader, Mohammed Salman Shazeeb, and Clifford Grant Lindsay. "Real-time object segmentation for laparoscopic cholecystectomy using YOLOv8." *Neural Computing and Applications* 37, no. 4 (2025): 2697–2710.
- [23] Casas, Edmundo, Leo Ramos, Cristian Romero, and Francklin Rivas-Echeverria. "A comparative study of YOLOv5 and YOLOv8 for corrosion segmentation tasks in metal surfaces." *Array* 22 (2024): 100351.
- [24] Mostafa, Ayman Mohamed, Alaa S. Alaerjan, Bader Aldughayfiq, Hisham Allahem, Alshimaa Abdelraof Mahmoud, Wael Said, Hosameldeen Shabana, and Mohamed Ezz. "Optimized YOLOv8 for enhanced breast tumor segmentation in ultrasound imaging." *Discover Oncology* 16, no. 1 (2025): 1152.
- [25] Alsabhan, Waleed, and Turkey Alotaiby. "Automatic building extraction on satellite images using Unet and ResNet50." *Computational Intelligence and Neuroscience* 2022, no. 1 (2022): 5008854.
- [26] Cheng, Jiayu, Yihang Zhu, Yiyang Zhao, Tong Li, Miaojin Chen, Qinan Sun, Qing Gu, and Xiaobin Zhang. "Application of an improved U-Net with image-to-image translation and transfer learning in peach orchard segmentation." *International Journal of Applied Earth Observation and Geoinformation* 130 (2024): 103871.
- [27] Zhu, Jiajun, Rui Zhang, and Haifei Zhang. "An MRI brain tumor segmentation method based on improved U-Net." *Mathematical biosciences and engineering: MBE* 21, no. 1 (2022): 778–791.
- [28] Saifullah, Shoffan, Anton Yudhana, and Andiko Putro Suryotomo. "Automatic Brain Tumor Segmentation: Advancing U-Net With ResNet50 Encoder for Precise Medical Image Analysis." *IEEE Access* 13 (2025): 43473–43489.
- [29] Sapkota, Ranjan, Dawood Ahmed, and Manoj Karkee. "Comparing YOLOv8 and Mask R-CNN for instance segmentation in complex orchard environments." *Artificial Intelligence in Agriculture* 13 (2024): 84–99.
- [30] Alsawaylimi, Amjad A. "Enhanced YOLOv8-Seg instance segmentation for real-time submerged debris detection." *IEEE Access* (2024).